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### Vertical cavity light-emitting element and manufacturing method of the same

#### Abstract

A vertical cavity light-emitting element includes a substrate, a first multilayer film reflecting mirror, a semiconductor structure layer, an electrode, an electrode layer, and a second multilayer film reflecting mirror. The first multilayer film reflecting mirror is formed on the substrate. The semiconductor structure layer includes a nitride semiconductor. The nitride semiconductor includes a first semiconductor layer that is formed on the first multilayer film reflecting mirror and is a first conductivity type, a second semiconductor layer that is formed on the first semiconductor layer and is the first conductivity type, a light-emitting layer that is formed on the second semiconductor layer and is configured to expose a region including an outer edge of a top surface of the second semiconductor layer, and a third semiconductor layer that is formed on the light-emitting layer and is a second conductivity type opposite to the first conductivity type. The electrode is formed on the top surface of the second semiconductor layer. The electrode layer is electrically in contact with the third semiconductor layer in one region of a top surface of the third semiconductor layer. The second multilayer film reflecting mirror constitutes a resonator with the first multilayer film reflecting mirror. The second semiconductor layer has a larger resistance than the first semiconductor layer.

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## Background/Summary

### BACKGROUND OF THE INVENTION

#### 1. Field of the Invention

(1) The present invention relates to a vertical cavity light-emitting element and a manufacturing method of the same.

#### 2. Background Art

(2) As a vertical cavity light-emitting element, for example, Japanese Patent No. 3763737 (JP-B2-2006-3763737) discloses a Vertical Cavity Surface Emitting Laser (VCSEL) that includes a semiconductor layer and two multilayer film reflecting mirrors mutually opposed across the semiconductor layer.

### SUMMARY OF THE INVENTION

(3) In the VCSEL as disclosed in Japanese Patent No. 3763737 (JP-B2-2006-3763737), it is one of the problems to flow a current flowing into the VCSEL to a region contributing to a gain of a laser beam in an active layer as much as possible, thus reducing a threshold current necessary for oscillating the laser beam as much as possible. It is also one of the problems to avoid an element breakdown due to an excessive current concentration in the semiconductor layer.

(4) The present invention has been made in consideration of the problems described above, and an object of the present invention is to provide a vertical cavity light-emitting element in which a threshold current is low and an element breakdown is less likely to occur, and a manufacturing method of the same.

(5) A vertical cavity light-emitting element according to the present invention includes a substrate, a first multilayer film reflecting mirror, a semiconductor structure layer, an electrode, an electrode layer, and a second multilayer film reflecting mirror. The first multilayer film reflecting mirror is formed on the substrate. The semiconductor structure layer includes a nitride semiconductor. The nitride semiconductor includes a first semiconductor layer that is formed on the first multilayer film reflecting mirror and is a first conductivity type, a second semiconductor layer that is formed on the first semiconductor layer and is the first conductivity type, a light-emitting layer that is formed on the second semiconductor layer and is configured to expose a region including an outer edge of a top surface of the second semiconductor layer, and a third semiconductor layer that is formed on the light-emitting layer and is a second conductivity type opposite to the first conductivity type. The electrode is formed on the top surface of the second semiconductor layer. The electrode layer is electrically in contact with the third semiconductor layer in one region of a top surface of the third semiconductor layer. The second multilayer film reflecting mirror constitutes a resonator with the first multilayer film reflecting mirror. The second semiconductor layer has a larger resistance than the first semiconductor layer.

(6) A manufacturing method of a vertical cavity light-emitting element according to the present invention includes: a first multilayer film reflecting mirror forming step of forming a first multilayer film reflecting mirror on a substrate; a semiconductor layer stacking step of stacking a GaN layer of a first conductivity type, an AlGaIn layer of the first conductivity type, a light-emitting layer, and a nitride semiconductor layer of a second conductivity type opposite to the first conductivity type on the first multilayer film reflecting mirror in this order; an etching process step of forming a mask in one region of a top surface of the nitride semiconductor layer and exposing a region including an outer edge of a top surface of the AlGaIn layer by performing an etching

process on another region excluding the one region of the top surface by a dry etching method while measuring an intensity of a reflected light when the other region is irradiated with a laser beam, the etching process step stopping the etching process based on a change of the intensity of the reflected light; an electrode forming step of forming an electrode on the top surface of the AlGaIn layer; an electrode layer forming step of forming an electrode layer electrically in contact with the nitride semiconductor layer in the one region of the top surface of the nitride semiconductor layer; and a second multilayer film reflecting mirror forming step of forming a second multilayer film reflecting mirror constituting a resonator with the first multilayer film reflecting mirror.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a vertical cavity surface emitting laser according to Embodiment 1;
- (2) FIG. 2 is a top view of the vertical cavity surface emitting laser according to Embodiment 1;
- (3) FIG. 3 is a cross-sectional view of the vertical cavity surface emitting laser according to Embodiment 1;
- (4) FIG. 4 is a flowchart of a manufacturing method of the vertical cavity surface emitting laser according to Embodiment 1;
- (5) FIG. 5 is a drawing illustrating a part of a manufacturing process of the vertical cavity surface emitting laser according to Embodiment 1;
- (6) FIG. 6 is a drawing illustrating a part of the manufacturing process of the vertical cavity surface emitting laser according to Embodiment 1;
- (7) FIG. 7 is a drawing illustrating a part of the manufacturing process of the vertical cavity surface emitting laser according to Embodiment 1;
- (8) FIG. 8 is a drawing illustrating a part of the manufacturing process of the vertical cavity surface emitting laser according to Embodiment 1; and
- (9) FIG. 9 is a top view of a vertical cavity surface emitting laser according to Embodiment 2.

### DETAILED DESCRIPTION OF THE INVENTION

- (10) The following describes embodiments of the present invention in detail with reference to the drawings.

#### Embodiment 1

- (11) FIG. 1 is a perspective view of a vertical cavity surface emitting laser (hereinafter simply referred to as a surface emitting laser) **10** according to Embodiment 1.
- (12) A substrate **11** is a flat plate-shaped substrate having a rectangular top surface shape. In the drawings of this application, as an example, a case where the substrate **11** has a square top surface shape is illustrated. The substrate **11** is a growth substrate that allows a growth of a semiconductor crystal on the top surface. In this embodiment, the substrate **11** contains gallium nitride (GaN).
- (13) A first multilayer film reflecting mirror **13** is a semiconductor multilayer film reflecting mirror in which a low refractive-index semiconductor film and a high refractive-index semiconductor film having a higher refractive index than the low refractive-index semiconductor film are alternately stacked on the substrate **11**. The first multilayer film reflecting mirror **13** is formed so as to cover the top surface of the substrate **11**. Therefore, the first multilayer film reflecting mirror **13** has a top surface shape the same as the top surface shape of the substrate **11**, that is, a rectangular top surface shape. In this embodiment, the first multilayer film reflecting mirror **13** is what is called a Distributed Bragg Reflector (DBR) containing a semiconductor material.
- (14) In this embodiment, the first multilayer film reflecting mirror **13** is formed by alternately stacking the low refractive-index semiconductor film having a composition of aluminum indium

nitride (AlInN) and the high refractive-index semiconductor film having a composition of GaN. For example, a buffer layer (not illustrated) having a composition of GaN is disposed on the top surface of the substrate **11**, and the high refractive-index semiconductor film and the low refractive-index semiconductor film described above are alternately formed on the buffer layer, thereby forming the first multilayer film reflecting mirror **13**.

(15) A semiconductor structure layer E M is a stacked structure including a plurality of semiconductor layers formed on the first multilayer film reflecting mirror **13**. In this embodiment, the semiconductor structure layer EM includes a first semiconductor layer **15**, a second semiconductor layer **16**, a third semiconductor layer **17**, an active layer **18**, and a fourth semiconductor layer **19**.

(16) The first semiconductor layer **15** is a nitride semiconductor layer that is formed on the first multilayer film reflecting mirror **13** and is an n-type as a first conductivity type. The first semiconductor layer **15** is formed so as to cover the top surface of the first multilayer film reflecting mirror **13**. Therefore, the first semiconductor layer **15** has a top surface shape the same as the top surface shape of the first multilayer film reflecting mirror **13**, that is, a rectangular top surface shape. In this embodiment, the first semiconductor layer **15** has a composition of GaN, and doped with silicon (Si) as n-type impurities.

(17) The second semiconductor layer **16** is a nitride semiconductor layer that is formed on the first semiconductor layer **15** and is the n-type. The second semiconductor layer **16** is formed so as to cover the top surface of the first semiconductor layer **15**. Therefore, the second semiconductor layer **16** has a top surface shape the same as the top surface shape of the first semiconductor layer **15**, that is, a rectangular top surface shape.

(18) In this embodiment, the second semiconductor layer **16** has a composition of aluminum gallium nitride (AlGaN), and doped with Si as n-type impurities. The second semiconductor layer **16** has a larger band gap energy than the first semiconductor layer **15** because of its composition.

(19) In this embodiment, the second semiconductor layer **16** is formed to have a thinner layer thickness than the first semiconductor layer **15**. Specifically, in this embodiment, the layer thickness of the first semiconductor layer **15** is 1400 nm, and the layer thickness of the second semiconductor layer **16** is 10 nm.

(20) The second semiconductor layer **16** is configured to have a higher resistance than the first semiconductor layer **15**. In the surface emitting laser **10**, the second semiconductor layer **16** is formed to have a higher resistance value of a material and the thinner layer thickness than the first semiconductor layer **15**. Accordingly, the second semiconductor layer **16** has a higher sheet resistance than the first semiconductor layer **15**.

(21) The third semiconductor layer **17** is a nitride semiconductor layer that is formed in the approximately center of a top surface **16T** of the second semiconductor layer **16** and is the n-type. The third semiconductor layer **17** has a circular top surface shape, and is formed to be configured to expose a region including an outer edge of the top surface **16T** of the second semiconductor layer **16**. In this embodiment, the third semiconductor layer **17** has a composition of GaN, and doped with Si as n-type impurities.

(22) The active layer **18** is a semiconductor layer formed so as to cover the top surface of the third semiconductor layer **17**. Therefore, the active layer **18** has a top surface shape the same as the top surface shape of the third semiconductor layer **17**, that is, a circular top surface shape.

(23) In this embodiment, the active layer **18** has a quantum well structure including a well layer having a composition of indium gallium nitride (InGaN) and a barrier layer having a composition of GaN. The active layer **18** is a layer configured to generate a light by recombination of the electrons and the holes in the active layer **18**. In other words, the active layer **18** is a light-emitting layer that emits a light from the active layer **18**.

(24) The fourth semiconductor layer **19** is a nitride semiconductor layer that is formed on the active layer **18** and is a p-type as a second conductivity type. The fourth semiconductor layer **19** is formed

so as to cover the top surface of the active layer **18**. Therefore, the fourth semiconductor layer **19** has a top surface shape the same as the top surface shape of the active layer **18**, that is, a circular top surface shape. In this embodiment, the fourth semiconductor layer **19** has a composition of GaN, and doped with magnesium (Mg) as p-type impurities.

(25) In this embodiment, a p-type AlGaIn layer (not illustrated) as a carrier block layer is formed between the active layer **18** and the fourth semiconductor layer **19**, and a p-type GaN layer (not illustrated) as a contact layer is formed on the fourth semiconductor layer **19**.

(26) In this embodiment, in the semiconductor structure layer EM, the third semiconductor layer **17**, the active layer **18**, and the fourth semiconductor layer **19** are stacked in this order on the top surface **16T** of the second semiconductor layer **16**, thereby forming a columnar portion extending upward from the top surface **16T** of the second semiconductor layer **16**. In other words, the semiconductor structure layer EM has a mesa-shaped structure (hereinafter also referred to as a mesa structure) including the third semiconductor layer **17** projecting from the top surface **16T** of the second semiconductor layer **16**, and the active layer **18** and the fourth semiconductor layer **19** are stacked on the mesa structure.

(27) An n-electrode NE is an annular metal electrode that is formed on the top surface **16T** of the second semiconductor layer **16** and electrically connected to the second semiconductor layer **16**. The n-electrode NE is formed so as to surround the third semiconductor layer **17** and to be spaced from the third semiconductor layer **17** on the top surface **16T** of the second semiconductor layer **16**. In this embodiment, the n-electrode NE is formed of titanium (Ti) and aluminum (Al) in this order on the top surface **16T** of the second semiconductor layer **16**.

(28) An insulating layer **21** is a translucent insulating layer including an insulator formed on the fourth semiconductor layer **19**. The insulating layer **21** is provided with a circular opening (not illustrated in FIG. 1) that exposes the fourth semiconductor layer **19** in the center of the insulating layer **21**. In this embodiment, the insulating layer **21** contains silicon dioxide (SiO<sub>2</sub>).

(29) A translucent electrode layer **23** is a conductive film that is formed on the insulating layer **21** so as to cover the above-described opening and has a translucency. The translucent electrode layer **23** has a circular top surface shape. In this embodiment, the translucent electrode layer **23** contains indium tin oxide (ITO).

(30) A p-electrode PE is an annular metal electrode that is formed along an outer edge of a top surface **23T** of the translucent electrode layer **23** and electrically connected to the translucent electrode layer **23**. In this embodiment, the p-electrode PE contains gold (Au).

(31) A second multilayer film reflecting mirror **25** is a dielectric multilayer film reflecting mirror in which a low refractive-index dielectric film and a high refractive-index dielectric film having a higher refractive index than the low refractive-index dielectric film are alternately stacked on the top surface **23T** of the translucent electrode layer **23**. The second multilayer film reflecting mirror **25** has a circular top surface shape smaller than the top surface of the translucent electrode layer **23**.

(32) In this embodiment, the second multilayer film reflecting mirror **25** is what is called a Distributed Bragg Reflector (DBR) containing a dielectric material. The second multilayer film reflecting mirror **25** is formed to be spaced from the p-electrode PE.

(33) In this embodiment, the second multilayer film reflecting mirror **25** is formed by alternately stacking the low refractive-index dielectric film containing SiO<sub>2</sub> and the high refractive-index dielectric film containing niobium oxide (Nb<sub>2</sub>O<sub>5</sub>). For the second multilayer film reflecting mirror **25**, tantalum pentoxide (Ta<sub>2</sub>O<sub>5</sub>) may be used as the low refractive-index dielectric film, and Al<sub>2</sub>O<sub>3</sub> may be used as the high refractive-index dielectric film.

(34) In this embodiment, the first multilayer film reflecting mirror **13** and the second multilayer film reflecting mirror **25** are disposed so as to interpose the semiconductor structure layer EM therebetween in an up-down direction in the drawing, and configured as a resonator to oscillate a light emitted from the active layer **18** of the semiconductor structure layer EM.

(35) In the surface emitting laser **10**, when a voltage is applied between the n-electrode NE and the

p-electrode PE, a current flows in the active layer **18** via a current confinement layer, and when a value of the current reaches a specific current value at which an intensity of an output light rapidly increases, that is, a threshold current, a light of stimulated emission is emitted from the active layer **18**. The light emitted from the active layer **18** is repeatedly reflected between the first multilayer film reflecting mirror **13** and the second multilayer film reflecting mirror **25**, thus reaching a resonant state (laser oscillation).

(36) In this embodiment, the first multilayer film reflecting mirror **13** has a reflectance slightly lower than that of the second multilayer film reflecting mirror **25**. Therefore, the light resonated between the first multilayer film reflecting mirror **13** and the second multilayer film reflecting mirror **25** is partially transmitted through the first multilayer film reflecting mirror **13** and the substrate **11**, and extracted outside. That is, the light resonated between the first multilayer film reflecting mirror **13** and the second multilayer film reflecting mirror **25** is emitted downward in the drawing

(37) In this embodiment, an anti-reflection film (not illustrated) in which Nb.sub.2O.sub.5 and SiO.sub.2 are stacked is formed on a lower surface of the substrate **11**. The anti-reflection film is what is called an AR coating that suppresses the reflection of the light emitted from the substrate **11** by the lower surface of the substrate **11**.

(38) FIG. 2 is a top view of the surface emitting laser **10**. In FIG. 2, an axis passing through the center of the surface emitting laser **10** in top view in a depth direction of the drawing is illustrated as a central axis CA. In this embodiment, the central axis CA corresponds to an optical axis of a laser beam emitted from the surface emitting laser **10**.

(39) As described above, the surface emitting laser **10** includes the semiconductor structure layer EM including the first semiconductor layer **15** and the second semiconductor layer **16** having the rectangular top surface shapes, and the third semiconductor layer **17**, the active layer **18**, and the fourth semiconductor layer **19** that are formed on the top surface **16T** of the second semiconductor layer **16** and have the circular top surface shapes.

(40) As described above, the surface emitting laser **10** includes the insulating layer **21** formed on the fourth semiconductor layer **19**, the translucent electrode layer **23** formed on the insulating layer **21**, and the second multilayer film reflecting mirror **25** formed on the top surface **23T** of the translucent electrode layer **23**.

(41) As described above, the insulating layer **21** is provided with a circular opening **210** that exposes the fourth semiconductor layer **19** in the center of the insulating layer **21**. In other words, the insulating layer **21** is an annular insulating layer formed on the fourth semiconductor layer **19**. As illustrated in FIG. 2, the opening **210** is covered with the second multilayer film reflecting mirror **25** in top view.

(42) As described above, the surface emitting laser **10** includes the n-electrode NE formed on the top surface **16T** of the second semiconductor layer **16** and the p-electrode PE formed on the top surface **23T** of the translucent electrode layer **23**. In top view, the n-electrode NE and the p-electrode PE have annular shapes that are mutually concentric circles with respect to the central axis CA, and the n-electrode NE is configured to have a larger diameter than the p-electrode PE.

(43) FIG. 3 is a cross-sectional view taken along a line 3-3 of the surface emitting laser **10** in FIG. 2. As described above, the surface emitting laser **10** includes the first multilayer film reflecting mirror **13** formed on the substrate **11**, the semiconductor structure layer EM formed on the first multilayer film reflecting mirror **13**, and the second multilayer film reflecting mirror **25** constituting a resonator with the first multilayer film reflecting mirror **13** while interposing the semiconductor structure layer EM therebetween.

(44) In this embodiment, the fourth semiconductor layer **19** includes a projecting portion **19P** that projects from the center of the fourth semiconductor layer **19** and has a circular top surface shape. For the projecting portion **19P**, a top surface **19PT** of the projecting portion **19P** is in contact with the lower surface of the translucent electrode layer **23**.

(45) The insulating layer **21** is provided with the opening **210** that exposes the top surface of the projecting portion **19P** in the top surface of the fourth semiconductor layer **19**. In other words, the insulating layer **21** is formed in an annular shape so as to surround the projecting portion **19P**. In this embodiment, the layer thickness of the insulating layer **21** is approximately the same as the layer thickness of the projecting portion **19P**.

(46) The translucent electrode layer **23** is formed so as to cover the top surface of the insulating layer **21** and the top surface of the projecting portion **19P** exposed from the opening **210** of the insulating layer **21**. That is, the translucent electrode layer **23** is electrically in contact with the fourth semiconductor layer **19** via the top surface **19PT** of the projecting portion **19P**. In other words, the translucent electrode layer **23** is electrically insulated from the fourth semiconductor layer **19** in a region not in contact with the top surface of the projecting portion **19P**.

(47) In this embodiment, the insulating layer **21** surrounding the projecting portion **19P** of the fourth semiconductor layer **19** is what is called a current confinement layer that restricts a range of current supply to the active layer **18**. That is, the current flowed from the p-electrode PE to the translucent electrode layer **23** flows in the active layer **18** via the top surface **19PT** of the projecting portion **19P** electrically in contact with the translucent electrode layer **23**.

(48) [Path of Current Flowing in Semiconductor Structure Layer]

(49) The following describes a path of the current flowing in the semiconductor structure layer EM in detail with reference to FIG. 3.

(50) In the surface emitting laser **10**, as described above, the second semiconductor layer **16** is formed to have the higher resistance value of the material and the thinner layer thickness than the first semiconductor layer **15**. Accordingly, the second semiconductor layer **16** has the higher sheet resistance than the first semiconductor layer **15**.

(51) According to this embodiment, with the above-described configuration of the surface emitting laser **10**, the current that has reached the top surface **19PT** of the projecting portion **19P** of the fourth semiconductor layer **19** from the p-electrode PE via the translucent electrode layer **23** flows in the first semiconductor layer **15** passing through a shortest path from the top surface **19PT** to the first semiconductor layer **15**, and flows to the n-electrode NE.

(52) Specifically, the current that has reached the top surface **19PT** of the projecting portion **19P** flows through each of the fourth semiconductor layer **19**, the active layer **18**, the third semiconductor layer **17**, and the second semiconductor layer **16** in the thickness direction, and flows to the first semiconductor layer **15**. The current that has reached the first semiconductor layer **15** is dispersed in an in-plane direction in the first semiconductor layer **15**, and flows to the n-electrode NE.

(53) In other words, in the flow of the current flowing in the semiconductor structure layer EM, a vector in a longitudinal direction is dominant in the fourth semiconductor layer **19**, the active layer **18**, the third semiconductor layer **17**, and the second semiconductor layer **16**, and a vector in a lateral direction is dominant in the first semiconductor layer **15**. Accordingly, the current flowing in from the top surface **19PT** flows mainly immediately downward from the top surface **19PT** until reaching the first semiconductor layer **15**, and flows in the lateral direction after entering the first semiconductor layer **15**.

(54) For example, as illustrated as a current path CP1 in FIG. 3, the current that has reached the top surface **19PT** of the projecting portion **19P** from the p-electrode PE flows to the first semiconductor layer **15** passing through mainly the central region of each of the fourth semiconductor layer **19**, the active layer **18**, the third semiconductor layer **17**, and the second semiconductor layer **16**.

(55) This embodiment allows avoiding the concentration of the current from the top surface **19PT** to a region RA (regions indicated by dashed lines in the drawing) that is a region adjacent to a rising portion of the mesa structure at the lower end of the third semiconductor layer **17**. Therefore, this embodiment allows avoiding an element breakdown and the like caused by the current concentration to the region RA.

(56) In this embodiment, the current flowing from the top surface **19PT** flows a region **LA** (region indicated by a two-dot chain line in the drawing) immediately below the projecting portion **19P** of the fourth semiconductor layer **19** in the active layer **18** in the thickness direction. The region **LA** is a region in which the injected current (carrier) contributes to a gain of a laser beam.

(57) Therefore, this embodiment allows avoiding a flow of the current to, for example, a region not contributing to the gain of the laser beam around the region **LA**, that is, a region in which an optical loss occurs. Therefore, since this embodiment allows avoiding the unnecessary flow of the current to the region in which an optical loss occurs, a threshold current necessary for oscillating the laser beam can be reduced.

(58) As described above, according to this embodiment, the element breakdown caused by the current concentration to the region **RA** and the like can be avoided, and the threshold current necessary for oscillating the laser beam can be reduced.

(59) Here, the following describes specific configurations of the respective layers in the semiconductor structure layer **EM** in the surface emitting laser **10**, which are preferable for producing the current flow that provides the effect of avoiding the element breakdown or the effect of reducing the threshold current as described above.

(60) In this embodiment, to provide the path in which the current flows in the in-plane direction in the first semiconductor layer **15**, the first semiconductor layer **15** preferably has the layer thickness ten times or more of that of the second semiconductor layer **16**. Specifically, the first semiconductor layer **15** preferably has the layer thickness of 500 nm or more and 1600 nm or less.

(61) In this embodiment, the second semiconductor layer **16** preferably has the layer thickness of 7 nm or more and 20 nm or less. This is because the layer thickness of less than 7 nm of the second semiconductor layer **16** decreases the distance between the third semiconductor layer **17** and the first semiconductor layer **15**, and it becomes difficult to avoid the current concentration to the region **RA**. Additionally, this is because the layer thickness exceeding 20 nm of the second semiconductor layer **16** possibly causes the increase of the voltage applied to the element.

(62) In this embodiment, the first semiconductor layer **15** containing GaN is preferably doped with Si having the concentration of from  $2 \times 10^{18}$  to  $5 \times 10^{18}$  atoms/cm<sup>3</sup> as an n-type dopant. This is because the Si concentration of less than  $2 \times 10^{18}$  atoms/cm<sup>3</sup> causes the current to be less likely to flow into the first semiconductor layer **15**, and the Si concentration exceeding  $5 \times 10^{18}$  atoms/cm<sup>3</sup> possibly increases an internal loss of the light due to a light absorption.

(63) In this embodiment, the second semiconductor layer **16** containing AlGaIn preferably has the Al content of 8% or more and 35% or less. This is because the Al content of less than 8% makes the current concentration to the region **RA** difficult to be avoided, and the Al content exceeding 35% increases a contact resistance between the second semiconductor layer **16** and the n-electrode **NE** formed on the top surface **16T** of the second semiconductor layer **16**.

(64) In this embodiment, the second semiconductor layer **16** containing AlGaIn is preferably doped with Si having the concentration of from  $3 \times 10^{18}$  to  $8 \times 10^{19}$  atoms/cm<sup>3</sup> as an n-type dopant. This is because the Si concentration of less than  $3 \times 10^{18}$  atoms/cm<sup>3</sup> increases a contact resistance between the n-electrode **NE** formed on the top surface **16T** of the second semiconductor layer **16** and the second semiconductor layer **16**, and the Si concentration exceeding  $8 \times 10^{19}$  atoms/cm<sup>3</sup> possibly deteriorates a crystallinity as the AlGaIn layer.

(65) In this embodiment, to reduce the internal loss of the light due to the laser beam absorption, the second semiconductor layer **16** is preferably formed at a position of a portion at which an optical electric field intensity decreases in the resonator, that is, a node part of the optical electric field distribution. The active layer **18** is preferably formed to have the center in the thickness direction at an antinode part of the optical electric field distribution.

(66) Specifically, the second semiconductor layer **16** is preferably formed such that the center in the thickness direction of the second semiconductor layer **16** is positioned at a position of a distance a

from the center in the thickness direction of the active layer **18**. Here, the distance  $a$  can be expressed as  $a=(2m+1)\times 0.25\lambda/n$ , and  $m$  is any given integer,  $\lambda$  is an oscillation wavelength of the laser beam, and  $n$  is a GaN refractive index at the oscillation wavelength.

(67) While the fourth semiconductor layer **19** as the contact layer is a p-type GaN layer in this embodiment, the fourth semiconductor layer **19** may be a p-type InGaN layer.

(68) [Manufacturing Method of Surface Emitting Laser]

(69) The following describes the manufacturing method of the surface emitting laser **10** with reference to FIG. 4. FIG. 4 is a flowchart illustrating an exemplary manufacturing process of the surface emitting laser **10**. In this embodiment, formations of the respective semiconductor layers in the semiconductor structure layer EM are performed by using a method of Metal Organic Chemical Vapor Deposition (MOCVD).

(70) First, as a step of forming the first multilayer film reflecting mirror **13**, trimethylaluminum (TMA) and trimethyl indium (TMIn), which are group III raw materials, and an ammonia gas are introduced on the substrate **11** containing GaN to form AlInN in MOCVD equipment.

Subsequently, trimethylgallium (TMGa) as a group III raw material is introduced instead of TMA and TMIn, thus forming GaN. This process is repeatedly performed **41** times, thus forming the first multilayer film reflecting mirror **13** containing **41** sets of AlInN/GaN (Step S101).

(71) At this time, AlInN and GaN are formed in layers on a (0001) plane of a crystal face of the substrate **11**, and formed so as to have the respective layer thicknesses of  $\frac{1}{4}$  of an optical layer thickness relative to a desired wavelength.

(72) Next, as a semiconductor layer stacking step, the first semiconductor layer **15** as a first n-type GaN layer, the second semiconductor layer **16** as an n-type AlGaIn layer, a second n-type GaN layer, an active layer, and a p-type GaN layer are stacked on the first multilayer film reflecting mirror **13** in this order (Step S102).

(73) As a specific stacking process, first, a doping gas SiH.sub.4 containing TMGa and Si is introduced, thus forming the first semiconductor layer **15** (first n-type GaN layer) having the layer thickness of 1400 nm and a Si doping amount of  $3\times 10^{18}$  atoms/cm<sup>3</sup>.

(74) Subsequently, raw materials of TMA are supplied on the first n-type GaN layer, thus forming the second semiconductor layer **16** (n-type AlGaIn layer) having the layer thickness of 10 nm and a Si doping amount of  $1\times 10^{19}$  atoms/cm<sup>3</sup> so as to have an Al composition of 20%.

(75) Subsequently, the second n-type GaN layer having the layer thickness of 100 nm and a Si doping amount of  $3\times 10^{18}$  atoms/cm<sup>3</sup> is formed on the second semiconductor layer **16**. The second n-type GaN layer is formed to reduce a lattice mismatch difference between the second semiconductor layer **16** and a non-doped InGaIn layer formed on the second n-type GaN layer.

(76) Subsequently, TMG and trimethyl indium (TMI) are supplied on the second n-type GaN layer, thus forming the non-doped InGaIn layer having an In composition of 5%. The non-doped InGaIn layer is formed to reduce a distortion of the active layer formed thereon. The InGaIn layer may be doped with Si in a range of from  $1\times 10^{17}$  to  $1\times 10^{18}$  atoms/cm<sup>3</sup> to be used as a layer for accelerating a carrier injection to the active layer.

(77) Subsequently, stacking a non-doped GaN barrier layer having a layer thickness of 4 nm and a non-doped InGaIn well layer having a layer thickness of 3 nm is repeatedly performed five times on the non-doped InGaIn layer, and then, a non-doped GaN barrier layer having a layer thickness of 5 nm is stacked, thus forming the active layer.

(78) Then, a Mg doped p-type AlGaIn carrier block layer (Al composition ratio 20%) having a layer thickness of 20 nm, a p-type GaN layer, and a p-type GaN contact layer are formed on the active layer in this order, thus ending the semiconductor layer stacking step.

(79) Next, an etching process is performed to the semiconductor layer stacked in Step S102 using a dry etching method (Step S103). Here, the step of performing the etching will be described in detail with reference to FIG. 5 to FIG. 8.

(80) FIG. 5 is a cross-sectional view of the surface emitting laser **10** in the middle of manufacturing

after stacking each of the semiconductor layers above the first multilayer film reflecting mirror **13** in Step **S102**. In the surface emitting laser **10** in the middle of manufacturing, the first semiconductor layer **15**, the second semiconductor layer **16**, a second n-type GaN layer **17M**, an active layer **18M**, and a p-type GaN layer **19M** are stacked on the first multilayer film reflecting mirror **13** in this order.

(81) FIG. **6** is a cross-sectional view of the surface emitting laser **10** in the middle of manufacturing after forming the projecting portion **19P** (current confinement layer in this embodiment) in the p-type GaN layer **19M**. The projecting portion **19P** is formed by removing a part of the p-type GaN layer **19M** using a known photolithography technique after Step **S102**. In this embodiment, the projecting portion **19P** is formed by forming a mask of  $\Phi 5.5\ \mu\text{m}$  in an approximately center region of a top surface of the p-type GaN layer **19M** and performing an etching process on a region excluding the mask formation region using an etching device.

(82) FIG. **7** is a cross-sectional view of the surface emitting laser **10** in the middle of manufacturing when the etching process on the top surface of the p-type GaN layer **19M** is started after forming the projecting portion **19P**. As illustrated in FIG. **7**, a mask **MS** is formed on the top surface of the p-type GaN layer **19M**. The mask **MS** is formed by applying a resist over the top surface of the p-type GaN layer **19M**, and removing a part of the resist so as to leave a concentric circle having a diameter larger than that of a portion corresponding to the projecting portion **19P**.

(83) An end-point detector **26** is a device for detecting a desired etching depth (end point) during the etching process of the semiconductor layer. For example, the end-point detector **26** is incorporated in the etching device. The end-point detector **26** includes a light projecting/receiving unit **27** and a controller **28**.

(84) The light projecting/receiving unit **27** is a light projecting/receiving unit including a light projecting unit that emits a laser beam and a light receiving unit that receives a reflected light when an object is irradiated with the laser beam. The controller **28** is connected to the light projecting/receiving unit **27**, and controls, for example, a position of the light projecting/receiving unit **27** and an irradiation intensity of the laser beam from the light projecting/receiving unit **27**.

(85) By using the light projecting/receiving unit **27** of the end-point detector **26**, the etching process is started on the region excluding the region in which the mask **MS** is formed of the top surface of the p-type GaN layer **19M** that is, the region in which the top surface of the p-type GaN layer **19M** is exposed while irradiating the region with the laser beam. In FIG. **7**, the light emitted from the light projecting/receiving unit **27** toward the top surface of the p-type GaN layer **19M** is indicated as a laser beam **L1**, and a light of a reflection of the laser beam **L1** is indicated as a reflected light **L2**.

(86) FIG. **8** is a cross-sectional view of the surface emitting laser **10** in the middle of manufacturing after exposing the region including an outer edge of the top surface of the second semiconductor layer **16** by the etching process. The configuration of the surface emitting laser **10** in the middle of manufacturing illustrated in FIG. **8** is obtained by removing each of the p-type GaN layer **19M**, the active layer **18M**, and the second n-type GaN layer **17M** excluding the region in which the mask is formed while measuring the intensity of the reflected light **L2** of the laser beam **L1** emitted from the light projecting/receiving unit **27**. At this time, the exposure of the second semiconductor layer **16** by the etching process can be determined based on the intensity of the reflected light **L2**.

(87) Specifically, the second semiconductor layer **16** can be determined to be exposed by detecting a difference between the intensity of the reflected light **L2** when the second n-type GaN layer **17M** formed on the second semiconductor layer **16** is irradiated with the laser beam **L1** and the intensity of the reflected light **L2** when the second semiconductor layer **16** is irradiated with the laser beam **L1** by using the light projecting/receiving unit **27** of the end-point detector **26**.

(88) The mask **MS** is removed after exposing the second semiconductor layer **16** by the etching process, thereby allowing the formation of the semiconductor structure layer **EM** having the mesa structure.

(89) Thus, by performing the etching process based on the intensity of the reflected light L2 of the laser beam L1 emitted to the semiconductor layer by using the end-point detector 26, the semiconductor structure layer EM including the second semiconductor layer 16 in which the region including the outer edge of the top surface is exposed and the mesa structure formed on the second semiconductor layer 16 can be formed.

(90) Refer to FIG. 4 again. After Step S103, Ti having a layer thickness of 10 nm and Al having a layer thickness of 500 nm are stacked in this order on the top surface of the second semiconductor layer 16 exposed in Step S103, thus forming the annular n-electrode NE (Step S104).

(91) Next, as a formation step of the insulating layer 21, a dielectric film containing SiO.sub.2 is disposed around the projecting portion 19P of the fourth semiconductor layer 19 so as to surround the projecting portion 19P (current confinement layer) of the fourth semiconductor layer 19 formed in Step S103 (Step S105). By Step S105, the top surface of the projecting portion 19P is exposed from the opening 210 of the insulating layer 21.

(92) Next, the translucent electrode layer 23 containing ITO is formed on the projecting portion 19P and the insulating layer 21 so as to cover the top surface of the projecting portion 19P exposed by Step S105 and the top surface of the insulating layer 21 (Step S106).

(93) Next, stacking Nb.sub.2O.sub.5 and SiO.sub.2 is repeatedly performed ten times on the translucent electrode layer 23 so as to cover the region above the projecting portion 19P in the top surface of the translucent electrode layer 23, thus forming the second multilayer film reflecting mirror 25 containing ten sets of Nb.sub.2O.sub.5/SiO.sub.2 (Step S107). At this time, the layers of Nb.sub.2O.sub.5 and SiO.sub.2 are formed so as to have the respective layer thicknesses of 1/4 of an optical layer thickness relative to a desired wavelength.

(94) Next, a layer of Au is formed along an outer edge of the top surface of the translucent electrode layer 23, thereby forming the annular p-electrode PE (Step S108).

(95) At last, an anti-reflection film (not illustrated) is formed on a backside surface (opposite surface of formation surface of first multilayer film reflecting mirror 13) of the substrate 11, thus manufacturing the surface emitting laser 10 illustrated in FIG. 3. The manufacturing process described above is merely an example, the elements doped to the semiconductor, the dimensions of the respective members, and the like can be changed as necessary.

## Embodiment 2

(96) Next, a surface emitting laser 20 according to Embodiment 2 will be described by referring to FIG. 9. The surface emitting laser 20 is different from Embodiment 1 in the configuration of the semiconductor structure layer EM, and similar to Embodiment 1 for other configurations.

(97) FIG. 9 is a cross-sectional view of the surface emitting laser 20 taken along the line 3-3 of FIG. 2 similarly to the cross-sectional view illustrated in FIG. 3. A fifth semiconductor layer 29 is a non-doped nitride semiconductor layer formed in contact with the lower surface of the second semiconductor layer 16 containing AlGaIn.

(98) The fifth semiconductor layer 29 is formed so as to cover the top surface of the first semiconductor layer 15. Therefore, the fifth semiconductor layer 29 has a top surface shape the same as the top surface shape of the first semiconductor layer 15, that is, a rectangular top surface shape. In this embodiment, the fifth semiconductor layer 29 has a composition of GaN.

(99) In this embodiment, an interface between the non-doped fifth semiconductor layer 29 containing GaN and the second semiconductor layer 16 containing AlGaIn is full of the electrons, and the interface has a higher conductivity than other regions.

(100) Specifically, in this embodiment, in a region in the fifth semiconductor layer 29 side in the interface between the fifth semiconductor layer 29 and the second semiconductor layer 16, a state where free electrons having high mobilities in the in-plane direction spread in a layer, that is, a two-dimensional electron gas layer is formed.

(101) In this embodiment, the current that has reached the top surface 19PT of the projecting portion 19P flows through each of the fourth semiconductor layer 19, the active layer 18, the third

semiconductor layer **17**, and the second semiconductor layer **16** in the thickness direction, and flows to the fifth semiconductor layer **29**. The current that has reached the fifth semiconductor layer **29** is dispersed in the in-plane direction in the two-dimensional electron gas layer in the fifth semiconductor layer **29**, and flows to the n-electrode NE.

(102) For example, as illustrated as a current path CP2 in FIG. **9**, the current that has reached the top surface **19PT** of the projecting portion **19P** from the p-electrode PE flows to the fifth semiconductor layer **29** passing through the center of each of the fourth semiconductor layer **19**, the active layer **18**, the third semiconductor layer **17**, and the second semiconductor layer **16**, and flows to the n-electrode NE passing through the second semiconductor layer **16** again in the thickness direction.

(103) Thus, according to this embodiment, since the current flows along the above-described path, a behavior of the current passing through the shortest path from the top surface **19PT** to the n-electrode NE, for example, a path in which the current flows from the top surface **19PT** and concentrates to the region RA in the proximity of the lower end of the third semiconductor layer **17** is avoided. In other words, this embodiment allows avoiding the concentration of the current to the proximity of the rising portion of the mesa structure. Therefore, this embodiment allows avoiding an element breakdown and the like caused by the current concentration to the region RA.

(104) In this embodiment, the current flowing from the top surface **19PT** flows a region LA (region in which the current contributes to the gain of the laser beam) immediately below the projecting portion **19P** of the fourth semiconductor layer **19** in the active layer **18** in the thickness direction similarly to Embodiment 1.

(105) Therefore, this embodiment allows avoiding a flow of the current to, for example, a region not contributing to the gain of the laser beam around the region LA, that is, a region in which an optical loss occurs. Therefore, since this embodiment allows avoiding the unnecessary flow of the current to the region in which an optical loss occurs, a threshold current necessary for oscillating the laser beam can be reduced.

(106) Thus, according to this embodiment, similarly to Embodiment 1, in the semiconductor structure layer EM of the surface emitting laser **10**, the threshold current necessary for oscillating the laser beam can be reduced while the current flows the region contributing to the gain of the laser beam in the semiconductor layer.

(107) The fifth semiconductor layer **29** preferably has a layer thickness of from 10 to 50 nm from an aspect of generating a two-dimensional electron gas between the second semiconductor layer **16** and the fifth semiconductor layer **29**. That is, the fifth semiconductor layer **29** has the thinner layer thickness than the first semiconductor layer **15**.

(108) In this embodiment, to configure the fifth semiconductor layer **29** as a path to disperse the current in the in-plane direction, the first semiconductor layer formed under the fifth semiconductor layer **29** may have a concentration of doped Si different from that of Embodiment 1. For example, the concentration of Si doped to the first semiconductor layer may be in a range of from  $5 \times 10^{17}$  to  $3 \times 10^{18}$  atoms/cm<sup>3</sup> as a concentration not causing an unstable operating voltage.

(109) Various numerical values, dimensions, materials, and the like in the embodiments and the modifications described above are merely examples, and can be appropriately selected depending on the usage and the surface emitting laser to be manufactured. For example, while the vertical cavity surface emitting laser is described in the embodiment, the application target of the configuration of the present invention is not limited to the surface emitting laser, and the configuration of the present invention is also applicable to other vertical cavity-type light-emitting devices, such as a vertical cavity light-emitting diode.

(110) In the embodiments and the modifications, ordinal numerals such as “first” and “second” are used for convenience, and do not indicate the stacking order of the members in some cases. Therefore, for example, the description may be given by replacing “first” with “second” as

necessary.

(111) It is understood that the foregoing description and accompanying drawings set forth the preferred embodiments of the present invention at the present time. Various modifications, additions and alternative designs will, of course, become apparent to those skilled in the art in light of the foregoing teachings without departing from the spirit and scope of the disclosed invention. Thus, it should be appreciated that the present invention is not limited to the disclosed Examples but may be practiced within the full scope of the appended claims. The present application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2021-94869 filed on Jun. 7, 2021, the entire contents of which are incorporated herein by reference.

## Claims

1. A vertical cavity light-emitting element comprising: a substrate; a first multilayer film reflecting mirror formed on the substrate; a semiconductor structure layer that includes a nitride semiconductor, the nitride semiconductor including a first semiconductor layer that is formed on the first multilayer film reflecting mirror and is a first conductivity type, a second semiconductor layer that is formed on the first semiconductor layer and is the first conductivity type, a light-emitting layer that is formed on the second semiconductor layer and is configured to expose a region including an outer edge of a top surface of the second semiconductor layer, and a third semiconductor layer that is formed on the light-emitting layer and is a second conductivity type opposite to the first conductivity type; an electrode formed on the top surface of the second semiconductor layer; an electrode layer electrically in contact with the third semiconductor layer in one region of a top surface of the third semiconductor layer; and a second multilayer film reflecting mirror constituting a resonator with the first multilayer film reflecting mirror, wherein the light-emitting layer has a quantum well structure including a well layer and a barrier layer, the first semiconductor layer has a composition of GaN, the second semiconductor layer has a composition of AlGaIn, a thickness of the second semiconductor layer being thinner than a thickness of the first semiconductor layer, and a resistance of the second semiconductor layer being larger than a resistance of the first semiconductor layer.
2. A manufacturing method of the vertical cavity light-emitting element according to claim 1, the method comprising: a first multilayer film reflecting mirror forming step of forming the first multilayer film reflecting mirror on the substrate; a semiconductor layer stacking step of stacking a GaN layer of the first conductivity type, an AlGaIn layer of the first conductivity type, the light-emitting layer, and a nitride semiconductor layer of the second conductivity type opposite to the first conductivity type on the first multilayer film reflecting mirror in this order; an etching process step of forming a mask in one region of a top surface of the nitride semiconductor layer and exposing a region including an outer edge of a top surface of the AlGaIn layer by performing an etching process on another region excluding the one region of the top surface by a dry etching method while measuring an intensity of a reflected light when the other region is irradiated with a laser beam, the etching process step stopping the etching process based on a change of the intensity of the reflected light; an electrode forming step of forming an electrode on the top surface of the AlGaIn layer; an electrode layer forming step of forming the electrode layer electrically in contact with the nitride semiconductor layer in the one region of the top surface of the nitride semiconductor layer; and a second multilayer film reflecting mirror forming step of forming the second multilayer film reflecting mirror constituting a resonator with the first multilayer film reflecting mirror.
3. The vertical cavity light-emitting element according to claim 1, wherein: the well layer has a composition of InGaIn, the barrier layer has a composition of GaN, and the third semiconductor layer has a composition of GaN.
4. The vertical cavity light-emitting element according to claim 1, wherein the thickness of the

second semiconductor layer is 7 nm or more and 20 nm or less.

5. The vertical cavity light-emitting element according to claim 1, wherein the thickness of the first semiconductor layer is ten times or more the thickness of the second semiconductor layer.

6. The vertical cavity light-emitting element according to claim 4, wherein the thickness of the first semiconductor layer is 500 nm or more and 1,600 nm or less.

7. The vertical cavity light-emitting element according to claim 1, wherein an Al content of the second semiconductor layer is 8% or more and 35% or less.

8. The vertical cavity light-emitting element according to claim 1, wherein, in a case in which  $m$  is any given integer,  $A$  is an oscillation wavelength of a laser beam, and  $n$  is a GaN refractive index at the oscillation wavelength, a distance  $a$  between a center of the second semiconductor layer in a thickness direction and a center of the light-emitting layer in a thickness direction is expressed as  $a = (2m + 1) \times 0.25\lambda/n$ .

9. A vertical cavity light-emitting element, comprising: a substrate; a first multilayer film reflecting mirror formed on the substrate; a semiconductor structure layer that includes a nitride semiconductor, the nitride semiconductor including a first semiconductor layer that is formed on the first multilayer film reflecting mirror and is a first conductivity type, a second semiconductor layer that is formed on the first semiconductor layer and is the first conductivity type, a light-emitting layer that is formed on the second semiconductor layer and is configured to expose a region including an outer edge of a top surface of the second semiconductor layer, and a third semiconductor layer that is formed on the light-emitting layer and is a second conductivity type opposite to the first conductivity type; an electrode formed on the top surface of the second semiconductor layer; an electrode layer electrically in contact with the third semiconductor layer in one region of a top surface of the third semiconductor layer; a second multilayer film reflecting mirror constituting a resonator with the first multilayer film reflecting mirror; and a fourth semiconductor layer formed in contact with a lower surface of the second semiconductor layer, the fourth semiconductor layer being non-doped and having a composition of GaN, wherein the first semiconductor layer has a composition of GaN, and the second semiconductor layer has a composition of AlGaIn, a resistance of the second semiconductor layer being larger than a resistance of the first semiconductor layer.

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